

CS401: Deep Learning and Neural Computation

Unit 1: Perceptrons and Activation Functions

1.1 Biological Neuron to Artificial Perceptron

An artificial neuron computes the weighted sum of its inputs plus a bias term, passed through a non-linear activation function.

$$z = \sum(w_i * x_i) + b = w^T * x + b$$

$$y_{\text{hat}} = \sigma(z)$$

Common Activation Functions:

1. Sigmoid: $\sigma(z) = 1 / (1 + \exp(-z))$, range (0, 1)
2. Hyperbolic Tangent (Tanh): $\tanh(z) = (\exp(z) - \exp(-z)) / (\exp(z) + \exp(-z))$, range (-1, 1)
3. Rectified Linear Unit (ReLU): $\text{ReLU}(z) = \max(0, z)$, solves vanishing gradient for positive activations
4. Leaky ReLU: $\text{LeakyReLU}(z) = \max(\alpha * z, z)$ where $\alpha = 0.01$

1.2 Loss Functions in Supervised Learning

For binary classification: Binary Cross-Entropy (BCE) Loss

$$L_{\text{BCE}} = - [y * \log(y_{\text{hat}}) + (1 - y) * \log(1 - y_{\text{hat}})]$$

For regression: Mean Squared Error (MSE) Loss

$$L_{\text{MSE}} = (1 / (2 * N)) * \sum((y_i - y_{\text{hat}_i})^2)$$

Unit 2: Optimization and Regularization

2.1 The Backpropagation Algorithm

Backpropagation computes partial derivatives of the loss L with respect to all network weights using the chain rule.

$$dL / dw_{ij} = (dL / dz_j) * (dz_j / dw_{ij}) = \delta_j * a_i$$

where δ_j is the error term of layer j .

Gradient Descent Weight Update Rule:

$$w_{\text{new}} = w_{\text{old}} - \eta * (dL / dw)$$

where η is the learning rate parameter.

2.2 Overfitting and Regularization Techniques

Overfitting occurs when a neural network achieves very low training error but fails to generalize to unseen test data.

Techniques to prevent overfitting:

1. L2 Regularization (Weight Decay): Adds penalty term $(\lambda / 2) * \sum(w_i^2)$ to loss function.
2. Dropout: Randomly sets a fraction p of hidden units to zero during each training forward pass.
3. Early Stopping: Halts training when validation loss stops improving for a specified patience threshold.
4. Data Augmentation: Synthesizes modified training examples via rotation, cropping, and noise injection.